SHEAFIFICATION IS REDUNDANT FOR SIMULTANEOUSLY CONTINUOUS AND COCONTINUOUS FUNCTORS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; every step the author writes out is inside the counted span and no closing sentence is a placeholder.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- The direct image of a sheaf along a continuous functor. Same chapter, Lemma lemma-pushforward-sheaf. Shows that no sheafification is needed for the direct image.
- The adjunction between the two functors on presheaves. Same chapter, Lemma lemma-adjoint-sheaves. Provides the left adjoint of the direct image on sheaves.
- Properties of adjoint functors. Categories, Section categories-section-adjoint. Supplies the statement about commutation with limits and colimits.

Source reference. Original lemma 00WW: sources/stacks-sites/pinned-original-source.tex, source lines 4818–4818. Navigation pointer invoked at proof line 4818; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 00WX: sources/stacks-sites/pinned-original-source.tex, source lines 4822–4822. Navigation pointer invoked at proof line 4822; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:sites.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original section 0036: sources/stacks-sites/pinned-original-source.tex, source lines 4824–4824. Navigation pointer invoked at proof line 4824; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $\mathcal{C}$ and $\mathcal{D}$ be sites. Let $u : \mathcal{C} \to \mathcal{D}$ be a functor. Assume that*

(a) *u is cocontinuous, and*
(b) *u is continuous.*

Let $g : Sh(\mathcal{C}) \to Sh(\mathcal{D})$ be the associated morphism of topoi. Then

(1) *sheafification in the formula $g^{-1} = (u^p\ )^\#$ is unnecessary, in other words $g^{-1}(\mathcal{G})(U) = \mathcal{G}(u(U))$,*
(2) *g^{-1} has a left adjoint $g_! = (u_p\ )^\#$, and*
(3) *g^{-1} commutes with arbitrary limits and colimits.*

Proof. By Lemma 00WW for any sheaf $\mathcal{G}$ on $\mathcal{D}$ the presheaf $u^p\mathcal{G}$ is a sheaf on $\mathcal{C}$ (and is denoted $u^s\mathcal{G}$). The functor $\mathcal{F} \mapsto (u_p\mathcal{F})^\#$ is a left adjoint to u^s by Lemma 00WX. The statement on limits and colimits follows from the discussion in Categories, Section 0036. □